ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

CONTROL SYSTEMS N6

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central N6 Control Systems N6 paper 1 · practice a.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct methods and clearly labelled sketches.

QUESTION 1

1.1 Summing junction with + reference input R(*s*) and − feedback ✓; error E(*s*) into G(*s*) giving output C(*s*) ✓; H(*s*) in the feedback path from output to summing junction ✓; all labels and arrows ✓ (4)

1.2 $\mathrm{C} = \mathrm{GE}$ ✓; $\mathrm{E} = \mathrm{R} - \mathrm{HC}$ ✓; $\mathrm{C} = \mathrm{G}(\mathrm{R} - \mathrm{HC})$, so $\mathrm{C}(1 + \mathrm{GH}) = \mathrm{GR}$ ✓; $\frac{\mathrm{C}}{\mathrm{R}} = \frac{\mathrm{G}}{1 + \mathrm{GH}}$ ✓ (4)

1.3 $\mathrm{T}(s) = \frac{10/s(s+2)}{1 + 10/s(s+2)}$ ✓ $= \frac{10}{s^2 + 2s + 10}$ ✓✓ (3)

1.4.1 $\mathrm{G} = \mathrm{G}_1\mathrm{G}_2 = \frac{10}{s+1}$ ✓✓; $\mathrm{T} = \frac{10/(s+1)}{1 + 0{,}1 \times 10/(s+1)}$ ✓ $= \frac{10}{s+2}$ ✓✓ (5)

1.4.2 $\mathrm{C}(s) = \frac{10}{s+2} \cdot \frac{1}{s}$ ✓; $c(\infty) = \lim_{s \to 0} s\mathrm{C}(s)$ ✓ $= \frac{10}{2}$ ✓ $= 5$ ✓ (4)

[20]

QUESTION 2

2.1.1 $\omega_n{}^2 = 10$ ✓; $\omega_n = 3{,}162$ rad/s ✓ (2)

2.1.2 $2\zeta\omega_n = 2$ ✓; $\zeta = \frac{1}{3{,}162} = 0{,}316$ ✓; $0 < \zeta < 1$, so underdamped ✓ (3)

2.1.3 $\omega_d = 3{,}162\sqrt{(1 - 0{,}316^2)}$ ✓ $= 3$ rad/s ✓ (2)

2.1.4 $\%\mathrm{OS} = e^{-0{,}316\pi/\sqrt{(1-0{,}1)}} \times 100$ ✓✓ $= 35{,}09\%$ ✓ (3)

2.1.5 $t_p = \frac{\pi}{3}$ ✓ $= 1{,}047$ s ✓ (2)

2.1.6 $t_s = \frac{4}{0{,}316 \times 3{,}162}$ ✓ $= 4$ s ✓ (2)

2.2 All curves start at zero and settle at the final value ✓; underdamped – overshoots and oscillates before settling ✓✓; critically damped – fastest response without overshoot ✓; overdamped – slow rise without overshoot ✓; axes labelled ✓ (6)

[20]

QUESTION 3

3.1 The type is the number of pure integrators (poles at the origin, factors of *s* in the denominator) in the open-loop transfer function ✓✓ (2)

3.2.1 Type 0 ✓ (1)

3.2.2 $K_p = \lim_{s\to 0} \frac{20}{(s+2)(s+5)} = \frac{20}{10} = 2$ ✓✓; $e_{ss} = \frac{1}{1+2}$ ✓ $= 0{,}333$ ✓ (4)

3.3 Type 1 system ✓; (a) $K_p = \infty$, so $e_{ss} = 0$ ✓✓ (b) $K_v = \lim_{s\to 0} s \cdot \frac{10}{s(s+2)} =$ 5 ✓; $e_{ss} = \frac{1}{5}$ ✓ $= 0{,}2$ ✓ (6)

3.4 Integral action adds a pole at the origin (raises the system type) ✓, which removes (or reduces) the steady-state error for a step input ✓ (2)

[15]

QUESTION 4

4.1.1 s^3: 1 11 ✓; s^2: 6 K ✓; s^1: $\frac{66-K}{6}$ 0 ✓✓; s^0: K ✓ (5)

4.1.2 For stability there must be no sign changes in the first column ✓; $\frac{66-K}{6} > 0$, so $K < 66$ ✓; $K > 0$ ✓; $0 < K < 66$ ✓ (4)

4.1.3 At $K = 66$, auxiliary equation $6s^2 + 66 = 0$ ✓; $s = \pm j\sqrt{11}$ ✓; $\omega = 3{,}317$ rad/s ✓ (3)

4.2 A stable system gives a bounded output for every bounded input / the response settles after a disturbance ✓✓; all closed-loop poles lie in the left half of the *s*-plane (negative real parts) ✓✓ (4)

4.3 Any TWO (2 marks each): smaller steady-state error ✓✓; faster response ✓✓; more overshoot / less damping; can make the system unstable if the gain is too high (4)

[20]

QUESTION 5

5.1 (a) Proportional – output is proportional to the error; sketch: a step in output ✓✓, but leaves an offset (steady-state error) ✓ (b) integral – output changes at a rate proportional to the error; sketch: a ramp ✓✓, removes offset but can cause overshoot ✓ (c) derivative – output is proportional to the rate of change of the error; sketch: a spike at the step ✓✓, improves stability / damping ✓ (9)

5.2 $K = \frac{100}{20}$ ✓ $= 5$ ✓ (2)

5.3 Derivative action responds only to a change in error; a constant error gives zero output ✓, so it cannot remove an error on its own and it amplifies noise ✓ (2)

5.4.1 Temperature rises to 183 °C, heater switches OFF; falls to 177 °C, heater switches ON – a sawtooth / cycling curve between 177 °C and 183 °C ✓✓; heater ON/OFF square wave drawn below in step with it ✓; axes and set point labelled ✓ (4)

5.4.2 It prevents the heater from switching on and off too rapidly (chattering) ✓, which reduces wear on the contactor / relay ✓ (2)

5.5 Any TWO: Ziegler–Nichols ultimate-gain (closed-loop) method ✓; Ziegler–Nichols process reaction curve (open-loop) method ✓; trial and error; auto-tuning by the controller (2)

5.6 Any FOUR: sensor / transmitter – measures the process variable (e.g. level) ✓; controller – compares the measured value with the set point and calculates the output ✓; final control element (e.g. control valve) – adjusts the manipulated variable ✓; process – the tank or system being controlled ✓; set point – the desired value (4)

[25]

TOTAL: 100

END OF MARKING GUIDELINE